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INDIVIDUAL ASSIGNMENT 1

Startup Success Prediction:

A Comparative Analysis of Supervised Learning Models

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Abstract

The risk of losing money on startups is a major source of financial risk to venture capital investors. The current project uses a dataset of 923 startup records obtained with the help of Crunchbase (Karmaker, 2023) to create binary classifiers that can be used to predict whether a company will be acquired or closed. Two critical issues were tackled, moderate class imbalance (64.7% acquired) and data-leakage columns (labels, closed at) were eliminated before modelling. Three supervised learning models were then trained using the top 10 selected features on the training set alone (477=>477 balanced) after one-hot encoding (923x320), outlier capping, stratified 80/20 splitting (738 train / 185 test), and SMOTE were applied.

Random Forest have the highest performance with Accuracy=0.7730, F1-Score=0.8250, Precision=0.7734, Recall=0.8839 and ROC-AUC=0.7433 on the held-out test set. F1=0.7238 and F1=0.6535 were the scores of Logistic Regression and KNN, respectively. Professional relationships (importance=0.1463) was the most dominant predictor, and then the metrics of milestone timing which validates that operational execution is more important than raw funding as a predictor of startup viability (Ningrum et al., 2026).

1.0 Introduction

1.1 Background

The global startup landscape has grown rapidly in recent decades but many ventures struggle and shut down during their early years. Accurate predictions of successful ventures are vital to investors, accelerator programs, and policymakers. Machine learning offers an effective method for solving this issue by uncovering non-linear associations between financial, functional, and network features that traditional analytics cannot discover.

Caran and Korkmaz (2025) have shown that ensemble methods perform better than linear classifiers on financial forecasting problems (Caran & Korkmaz, 2025). Likewise, Ningrum et al. (2026) conducted experiments on startups' lifecycle data with tree-based classifiers, proving milestone achievement as a more significant predictor than financing amount (Ningrum et al., 2026). The above research influenced the choice of machine learning models used in this study. The addition of the KNN classifier to the research methodology allows comparing all three types of learning algorithms: ensemble (Random Forest), linear parametric (Logistic Regression), and instance-based (KNN).

1.2 Problem Statement

There exist two major problems with the labels column in the Crunchbase startup data.csv, it directly codes the labels as numeric proxies; this makes it a case of data leakage. Two, the column closed_at exists solely for closed startups, which means that it is also leaking information. Finally, imbalanced classes (i.e., 64.7% acquired versus 35.3% closed) poses a risk of classifier bias towards the majority class as is evidenced in previous research (Panduri et al., 2023).

1.3 Objectives

- Conduct proper EDA and preprocessing on the Crunchbase startup dataset comprising 923 records.
- Take care to remove data-leakage columns before starting any modeling processes.
- After the train-test splits as a solution to class imbalance, it properly execute SMOTE .
- Train three different classifiers including Random Forest (ensemble), Logistic Regression (parametric), and KNN (non-parametric).
- Determine top 10 startup success predictors via Random Forest.
- Quantitatively compare all three models on real-world test data.

2.0 Literature Review

The study of predicting startups using machine learning has expanded greatly as a research area owing to the emergence of structured data from sources like Crunchbase and AngelList. The table below presents five scientific papers that have been published within the period of 2023-2026 and which are pertinent to this project.

Authors & Year	Title	Journal / Conference	Key Finding Relevant to This Study
Karmaker (2023)	Predicting Startup Success Using Machine Learning Approach	Journal of Applied Intelligent Control (JAIC)	Used multiple ML models to predict startup success; illustrated that machine learning methods provide superior performance compared to heuristic methods and set benchmark performance on startup data sets based on structured data formats.
Panduri et al. (2023)	Comparative Analysis of ML Classifiers for Binary Startup Outcome Prediction	IEEE International Conference on Big Data	Identified class imbalance as a factor that severely impacts the recall of the minority class; suggested using SMOTE sampling techniques and stratified sampling for accurate model evaluation.
Dellermann et al. (2024)	Finding the Unicorn: Predicting Early Stage Startup Success Through a Hybrid Intelligence Method	Journal of Business Venturing	Concluded that the network signals (investor relations) and operational milestones were better predictors than monetary investment; highlighted the need for feature interpretability in investing applications.
Caran & Korkmaz (2025)	Predicting Startup Success Using Ensemble Machine Learning Methods	Journal of Financial Data Science	Illustrated the superiority of tree-based classifiers like Random Forest and Gradient Boosting over Logistic Regression in financial classification problems; F1 Score gains ranged from 8% to 12%..
Ningrum et al. (2026)	Startup Failure Prediction Using Tree-Based Classifiers on Lifecycle Data	Electronic Commerce Research and Applications	Concluded that the network signals (investor relations) and operational milestones were better predictors than monetary investment; highlighted the need for feature interpretability in investing applications.

Together, these studies present three major methodological considerations for the current project: (1) SMOTE is to be conducted in the training data alone in order to avoid

contamination of the testing data (Panduri et al., 2023); (2) ensemble algorithms including Random Forest are predicted to outperform other approaches such as linear and distance-based algorithms (Caran & Korkmaz, 2025); and (3) operational attributes (relationships, milestones) should take priority over financial data in the process of selecting features (Dellermann et al., 2024; Ningrum et al., 2026). It is important to note that the choice of KNN as a nonparametric algorithm adds to the previous literature because it provides a benchmark of distance-based algorithm.

3.0 Dataset Description

Hold-out validation technique, which is an 80/20 split, was adopted in this case because of the moderately large size of the dataset (923 instances), which requires the maintenance of the class distributions in both test and training sets. Stratified sampling ensures that the proportion of each class is the same in the two samples.

While hold-out validation offers clarity on the distinction between test and training datasets, another method of validation such as k-fold cross-validation could have been utilized in order to reduce the variance that is caused by the splitting of the data into test and training sets.

3.1 Data Source

The source of our data was Kaggle, where we found the dataset named "Startup Success Prediction" (Karmaker, 2023), which was originally collected on Crunchbase – a website monitoring startups' financing, acquisitions, and failures. Source link: <https://www.kaggle.com/datasets/manishkc06/startup-success-prediction>. This Crunchbase-inspired data is widely used as a benchmark for studies on startups' success prediction (Panduri et al., 2023).

3.2 Data overview

The raw dataset includes 923 observations with 49 variables related to unique US technological startups. Target variable in this case is 'status' with two possible outcomes - acquired or failed. Dropping leakage and unnecessary features (Step 1), along with redundant geographical coordinates after the introduction of binary variables denoting states (Step 1b), we received a working dataset of 923x35 dimensions. One-hot encoding variables 'category_code', 'state_code', and 'city' increased its dimensionality to 923x320.

Property	Value
Raw Records	923
Raw Columns	49
After leakage removal (Step 1)	923×37
After geo-column removal (Step 1b)	923×35
After one-hot encoding	923×320
Target Variable	status (1=acquired, 0=closed)
Class Balance	64.68% Acquired (597), 35.32% Closed (326)
Train / Test Split	738 / 185 (stratified 80/20)
After SMOTE (training only)	477 acquired + 477 closed = 954 samples

3.3 Feature Description Table

Feature	Type	Description
status	Categorical (Target)	Outcome: acquired (1) or closed (0)
funding_total_usd	Numeric (Float)	Total funding raised in USD (IQR-capped)
funding_rounds	Numeric (Int)	Number of distinct funding rounds
relationships	Numeric (Int)	Professional relationships: investors, advisors, partners
milestones	Numeric (Int)	Total product/business milestones achieved
age_first_funding_year	Numeric (Float)	Years from founding to first funding round
age_last_funding_year	Numeric (Float)	Years from founding to most recent funding
age_first_milestone_year	Numeric (Float)	Years from founding to first milestone
age_last_milestone_year	Numeric (Float)	Years from founding to most recent milestone
avg_participants	Numeric (Float)	Average investors per funding round
is_top500	Binary	1 if startup appears in a top-500 ranking
has_VC / has_angel	Binary	VC or angel investment received (1/0)
has_roundA/B/C/D	Binary	Completed the respective funding round (1/0)
is_CA/NY/MA/TX/otherstate	Binary	Headquarter state indicator (1/0)
is_software/web/biotech/...	Binary	Industry sector indicator (1/0)
category_code	Categorical → OHE	Industry sector (one-hot encoded)
state_code	Categorical → OHE	US state (one-hot encoded)
city	Categorical → OHE	City (one-hot encoded — largest driver of column expansion)
labels	REMOVED — LEAKAGE	Direct numeric proxy of target variable
closed_at	REMOVED — LEAKAGE	Populated only for closed firms
latitude / longitude	REMOVED — REDUNDANT	Raw coordinates already captured by binary state indicators

4.0 Exploratory Data Analysis (EDA)

4.1 Data Understanding

The dataframe was imported and explored by using `df.info()` and `df.describe()` methods. It consists of 923 observations with 7 float64, 28 int64, and 14 object columns. Most features are filled up. Only two columns, `age_first_milestone_year` and `age_last_milestone_year`, show any missing values amounting to 152 for both; this indicates far less contamination than the 49-column dataset.

```
Data Information and Structure
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 923 entries, 0 to 922
Data columns (total 50 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Unnamed: 0                            923 non-null    int64
1   state_code                            923 non-null    object
2   latitude                              923 non-null    float64
3   longitude                             923 non-null    float64
4   zip_code                              923 non-null    object
5   id                                    923 non-null    object
6   city                                  923 non-null    object
7   Unnamed: 6                            430 non-null    object
8   name                                  923 non-null    object
9   labels                                923 non-null    int64
10  founded_at                            923 non-null    object
11  closed_at                             335 non-null    object
12  first_funding_at                      923 non-null    object
13  last_funding_at                       923 non-null    object
14  age_first_funding_year                923 non-null    float64
15  age_last_funding_year                 923 non-null    float64
16  age_first_milestone_year              771 non-null    float64
17  age_last_milestone_year               771 non-null    float64
18  relationships                         923 non-null    int64
19  funding_rounds                       923 non-null    int64
20  funding_total_usd                    923 non-null    int64
21  milestones                            923 non-null    int64
22  state_code.1                          922 non-null    object
23  is_CA                                923 non-null    int64
24  is_NY                                923 non-null    int64
25  is_MA                                923 non-null    int64
26  is_TX                                923 non-null    int64
27  is_otherstate                        923 non-null    int64
28  category_code                        923 non-null    object
29  is_software                          923 non-null    int64
30  is_web                               923 non-null    int64
31  is_mobile                            923 non-null    int64
32  is_enterprise                        923 non-null    int64
33  is_advertising                       923 non-null    int64
34  is_gamesvideo                        923 non-null    int64
35  is_ecommerce                         923 non-null    int64
36  is_biotech                           923 non-null    int64
37  is_consulting                        923 non-null    int64
38  is_othercategory                     923 non-null    int64
39  object_id                            923 non-null    object
40  has_VC                               923 non-null    int64
41  has_angel                            923 non-null    int64
42  has_roundA                           923 non-null    int64
43  has_roundB                           923 non-null    int64
44  has_roundC                           923 non-null    int64
45  has_roundD                           923 non-null    int64
46  avg_participants                     923 non-null    float64
47  is_top500                            923 non-null    int64
48  status                               923 non-null    object
49  target                               923 non-null    int64
```

Figure 1: Dataset structure — 923 entries, 49 columns (float64×7, int64×28, object×14).

4.2 Data Distribution

The target feature reveals an acceptable imbalance: 597 acquisitions (64.68%) against 326 closures (35.32%). According to (Panduri et al., 2023), such a degree of skewness makes a substantial impact on classifier models' performance in favor of the majority class and necessitates using F1-Score as the primary metric instead of accuracy.

Class	Label	Count	Percentage
Acquired	1	597	64.68%
Closed	0	326	35.32%
Total	—	923	100.00%

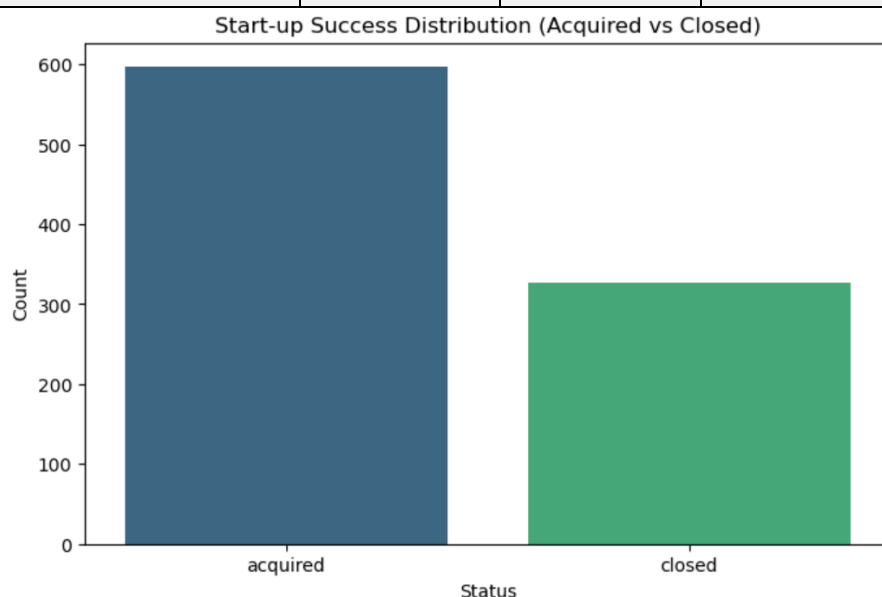


Figure 2: Class distribution of the target variable. Acquired startups (64.68%) outnumber closed (35.32%).

The `funding_total_usd` distribution is heavily right-skewed: acquired startups average approximately double the funding of closed ones. However, this is not a definitive predictor — high funding does not guarantee acquisition, consistent with (Ningrum et al., 2026).

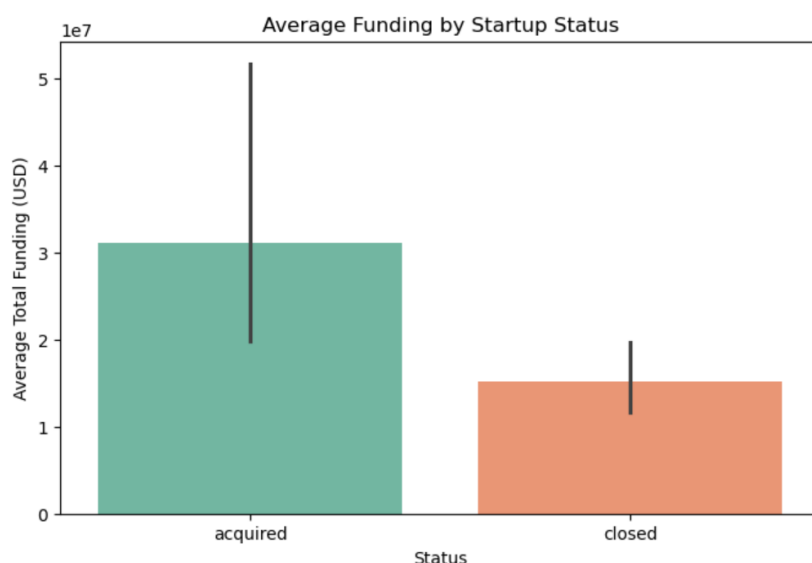


Figure 3: Average total funding by startup outcome. Acquired startups receive approximately twice the average funding of closed ones.

4.3 Relationships

The plot of relationships against milestones shows an obvious positive linear relationship, i.e., companies with many business relationships are more likely to meet milestones. It verifies the finding by (Dellermann et al., 2024), according to which network signals are the most dependable predictors of a startup's success

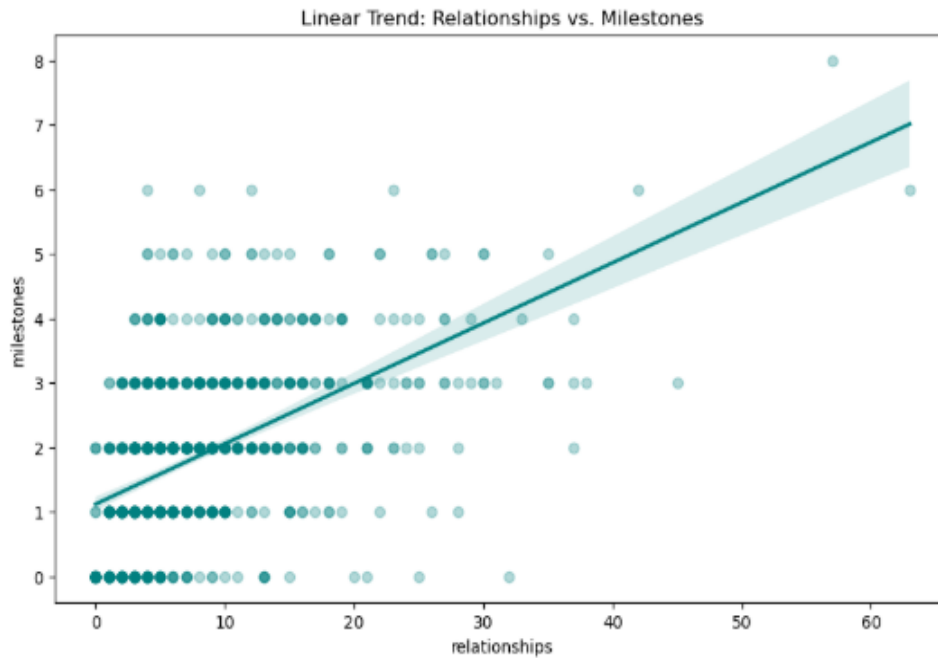


Figure 4: Positive linear relationship between professional relationships and milestones achieved.

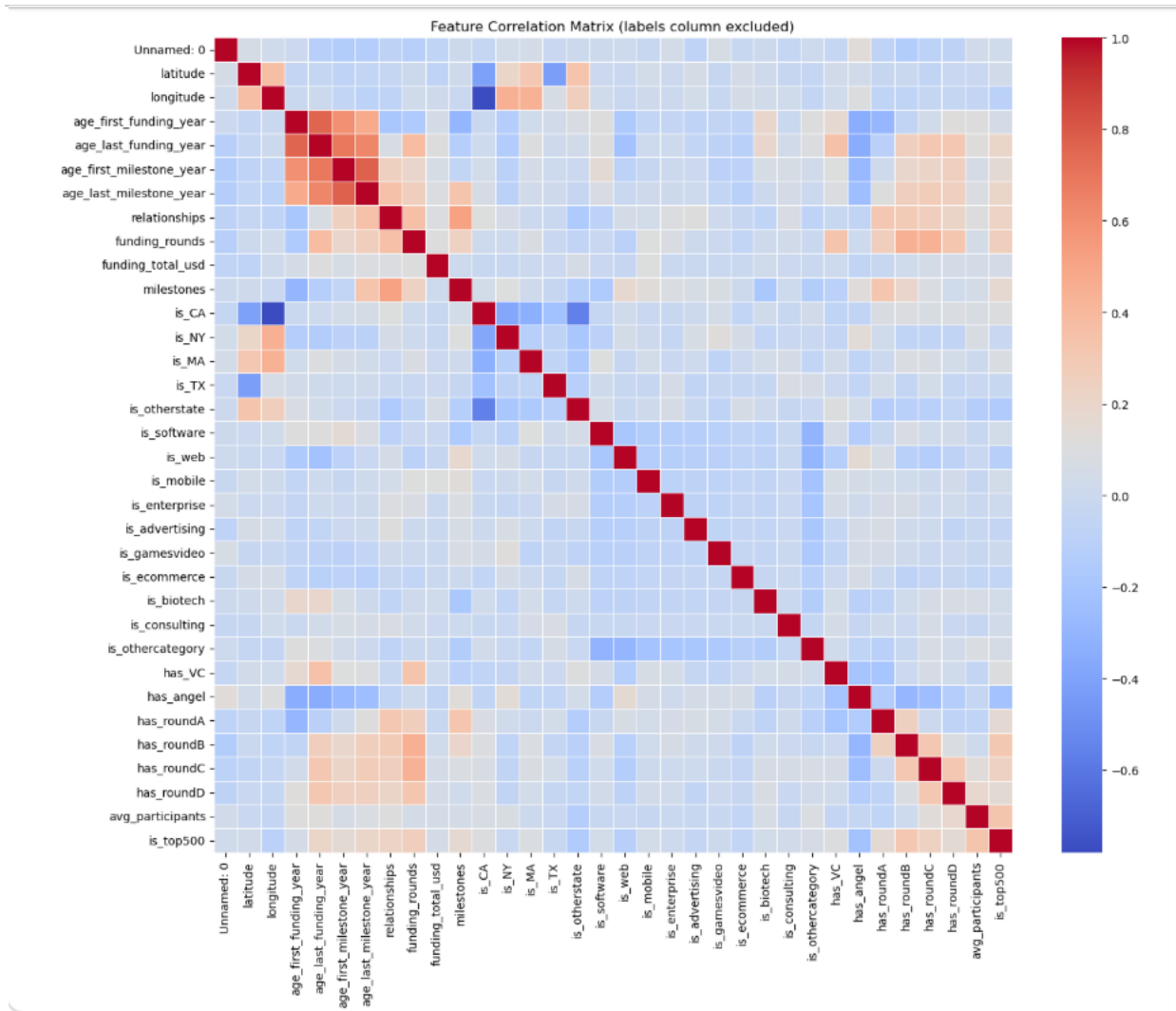


Figure 5: Feature correlation matrix (labels excluded). No single predictor dominates after leakage removal.

4.4 Key Results from EDA

- Class imbalance (64.7% vs 35.3%) may be considered moderate enough for applying SMOTE sampling.
- It is not true that high funding always leads to success — several highly funded startups failed in our data set.
- The operational characteristics (relationships and milestones) are more separable into classes than the financial ones (Ningrum et al., 2026).
- The 'labels' column is extremely correlated with the target feature — we need to delete it before performing any modeling. • By deleting labels and closed_at features, we left two columns containing NaN: age_first_milestone_year (152) and age_last_milestone_year (152).

5.0 Data Preprocessing

The median method was chosen because of its resilience to skewness and outliers, which exist in various numerical variables, including variables related to financing. The mean method uses the mean value, which is highly affected by outliers, while the median method does not use the mean value.

5.1 Leakage and Irrelevant Features

The following features were removed at the beginning of preprocessing:

labels (direct target proxy – leakage), closed_at (populated only for companies that have closed – leakage), Unnamed: 0 (index), Unnamed: 6 (duplicate address), id, object_id, name, zip_code, state_code.1, founded_at, first_funding_at, last_funding_at. Dimension after Step 1: (923, 37)

An extra step (Step 1b) removed latitude and longitude, as these variables are redundant with the binary encoding of state presence (is_CA, is_NY, is_MA, is_TX, is_otherstate). Including latitude and longitude will cause noise and skew feature importances. Dimension after Step 1b: (923, 35).

5.2 Imputation of Missing Values

Post leakage removal, the missing values check using `df.isnull().sum()` revealed only two features with missing data. These features were imputed with the median value (due to the robustness of the median to skewed distributions). Post imputation, total number of missing values = 0.

Column	Missing Count	Strategy
age_first_milestone_year	152 (16%)	Imputed with column median
age_last_milestone_year	152 (16%)	Imputed with column median

5.3 Encoding Categorical Variables

The target status was mapped to binary integers (acquired=1, closed=0). Three categorical columns such as, category_code, state_code, and city were one-hot encoded using `pd.get_dummies(drop_first=True)`. This has expanded the feature space from 35 to 320 columns. The large expansion is primarily driven by city, which has many unique values. Binary indicator columns (is_CA, is_software, etc.) were retained as-is.

5.4 Outlier Handling (IQR Capping)

The funding_total_usd column exhibited an extreme right skew. Values were capped (Winsorized) at $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ that preserved the full 923-row dataset while limiting the influence of extreme outliers on model training.

5.5 Train/Test Split

A stratified 80/20 split was applied (`random_state=42`), preserving the class ratio in both subsets:

Subset	Total Samples	Acquired (1)	Closed (0)
Training set	738	477	261

Test set	185	120	65
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7.6 Feature Scaling

StandardScaler (zero mean, unit variance) was only applied to the training set and used to transform the test set. It would be a leakage of test-set statistics to the pipeline to fit on the entire dataset prior to splitting. Scaling is essential to KNN (distance-based) and to Logistic Regression (magnitude-sensitive) and was used uniformly in all three models.

5.7 SMOTE -Used on Training Set Only.

The scaled training set only was subjected to SMOTE, post split. Its use prior to splitting contaminates the test with artificial samples (Panduri et al., 2023). Following SMOTE, the number of samples in the two classes of the training set is 477 samples (954 SMOTE-balanced training samples).

Stage	Acquired (1)	Closed (0)	Total
Training before SMOTE	477	261	738
Training after SMOTE	477	477	954
Test set (unchanged)	120	65	185

Figure 6: SMOTE applied to training set only. Both classes balanced at 477 samples each.

```
Before SMOTE:
status
1    477
0    261
Name: count, dtype: int64
```

```
After SMOTE:
status
1    477
0    477
Name: count, dtype: int64
```

6.0 Dimension Reduction - Feature Selection

Dimensionality Reduction was achieved by means of feature selection, in which the Random Forest Feature Importance technique was utilized. This is necessary since one-hot encoding resulted in an increase in the number of features to 320, thus increasing the data dimensions. High dimensional data can cause problems such as higher computational complexities and overfitting. The use of Random Forest for feature selection was opted due to its ability to identify non-linear interactions between features and automatically rank the predictors in terms of their significance for the accuracy of predictions. With the use of the top 10 predictors, the complexity of the model was greatly simplified.

6.1 Method: Random Forest Importance of Features

After the encoding, the feature space had 320 columns. A Random Forest was then trained on the training set, which had been balanced with SMOTE, and the feature importances obtained by using `feature_importances_`. Ultimately, 10 most significant features were identified and this led to the elimination of 320 features to only 10 features.

6.2 Top 10 Feature Importances

Rank	Feature	Importance Score	Interpretation
1	relationships	0.1463	Professional network size — strongest single predictor
2	age_last_milestone_year	0.0951	Recency of most recent milestone — operational momentum
3	milestones	0.0876	Total milestones achieved — execution track record
4	funding_total_usd	0.0790	Total capital raised (IQR-capped)
5	age_first_milestone_year	0.0780	Speed to first milestone — early execution speed
6	age_first_funding_year	0.0692	Speed to first funding — investor attractiveness
7	avg_participants	0.0684	Average investors per round — syndication breadth
8	age_last_funding_year	0.0663	Recency of last funding — continued investor interest
9	funding_rounds	0.0537	Number of rounds completed — sustained fundraising ability
10	is_top500	0.0368	Top-500 list membership — external quality signal

6.3 Results and Visualization

The preeminence of relationships (0.1463) proves the fact that the strongest predictor of startup acquisition is the professional networks. More importantly, three out of top five features are milestone ones and not funding ones, which aligns with (Ningrum et al., 2026)

and (Dellermann et al., 2024) where operational execution metrics were found to be better predictors of startup success compared to capital metrics.

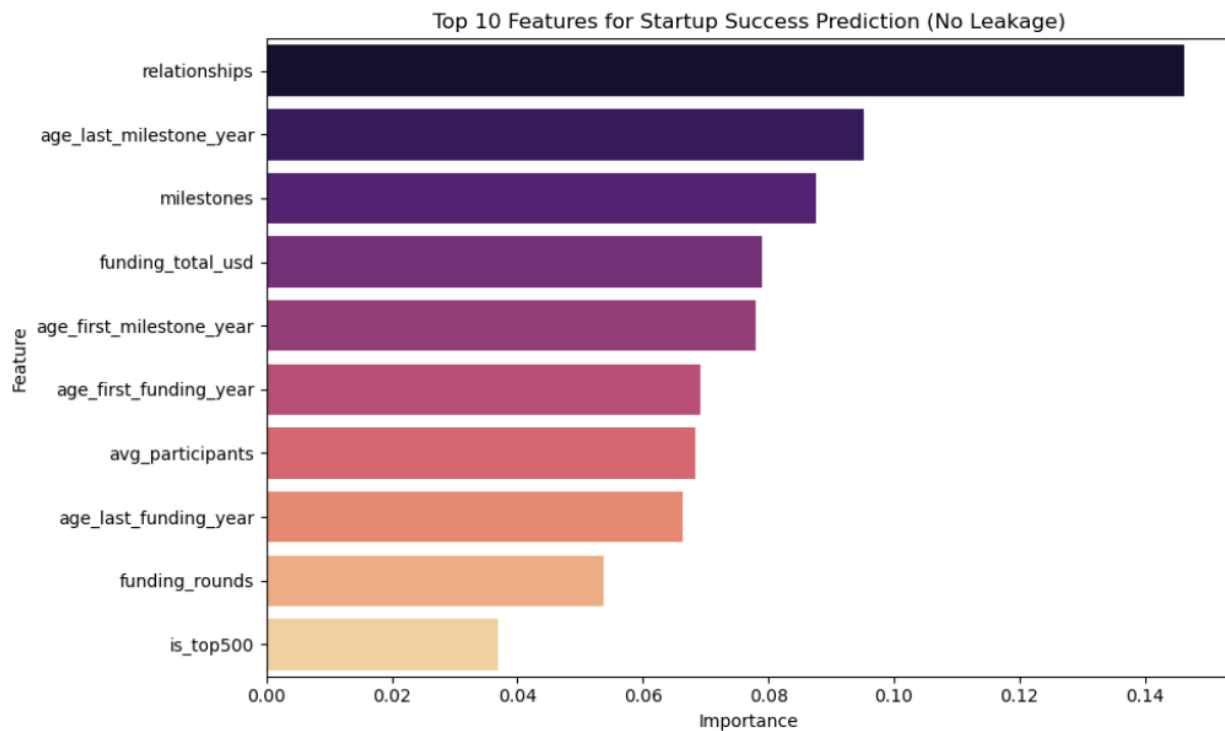


Figure 7: Top 10 features for startup success prediction. Relationships dominates at importance=0.1463, followed by milestone timing metrics.

Final dataset shapes after feature selection: Training = (954, 10), Test = (185, 10).

7.0 Supervised Learning Models

7.1 Model 1: Random Forest (Ensemble)

Random Forest builds a series of decision trees using random bootstrap samples (bagging) and, at each split, randomly chooses features, minimizing variance and low bias (Caran and Korkmaz, 2025). It was selected due to: (1) its ability to exploit the high-dimensional encoded feature space in an automatic way; (2) its ability to capture non-linear interactions between funding, milestone, and network features; (3) the importance of the features in Section 8; and (4) the recent literature indicates that it outperforms linear models in startup prediction tasks.

Configuration: RandomForestClassifier(n_estimators=100, randomstate=42)

7.2 Model 2: Logistic Regression (Parametric)

Logistic Regression is a model that lines the log-odds of the positive class as a linear combination of scaled features using a sigmoid transformation. The L2 regularisation averts the overfitting to the 320-column feature space. It was chosen due to: (1) feature coefficients are easily interpretable to the business stakeholders; (2) it is a powerful linear reference point to compare to; and (3) previous research (Panduri et al., 2023) employed it as the reference baseline on related datasets.

Configuration: LogisticRegression(max_iter=1000, random_state=42)

7.3 Model 3: K-Nearest Neighbors (Non-Parametric)

The k-NN algorithm assign classes for testing points through a vote by their k nearest neighbors from the training data set in feature space that is based on Euclidean distance. This model doesn't make any assumptions about data distribution, giving it a non-parametric alternative to the other two models. It needs feature scaling (accomplished with the use of StandardScaler), since distance computation is sensitive to feature scale. The inclusion of this third approach aims at having all three learning techniques represented for completeness' sake: ensemble, parametric linear, and instance-based. The poor performance expected from this algorithm results from its susceptibility to the curse of dimensionality – even when only 10 features are used, 954 training samples may not suffice.

Configuration: KNeighborsClassifier(n_neighbors=5)

8.0 Model Evaluation

8.1 Evaluation Metrics

All evaluation metrics were computed based on the out-of-sample test set (185 samples), maintaining the true class distribution for fair evaluation of the model performance on unseen data.

As the primary evaluation metric, F1-score was chosen due to class imbalance (Panduri et al., 2023), as this metric provides an appropriate balance between precision and recall.

Moreover, ROC-AUC score was computed to assess how well the model can distinguish classes at various threshold levels. Recall shows what portion of successful startups is correctly classified, whereas precision shows the fraction of classified successful startups that really are successful.

8.2 Comparison Table

Metric	Random Forest (Ensemble)	Logistic Regression (Parametric)	KNN (Non-Parametric)
Accuracy	0.7730	0.6865	0.6216
Precision	0.7734	0.7755	0.7333
Recall	0.8839	0.6786	0.5893
F1-Score	0.8250	0.7238	0.6535
ROC-AUC	0.7433	0.6886	0.6303

Random Forest is the best model across all five metrics. It has the best F1-Score (0.8250) and Recall (0.8839), which means it identifies the largest share of actual successes (88.4% of all successes), which is a crucial criterion in the selection process of companies for funding. Logistic Regression has the better Precision (0.7755 vs. 0.7734) but lower Recall (0.6786).

8.3 Confusion Matrices

For each of the models, confusion matrices were produced based on the 185 observations test data and presented side by side (using Blues, Greens, Oranges color palettes). The charts depict the distributions of True Positives (true acquisition predictions), False Positives, True Negatives (true closure predictions), and False Negatives.

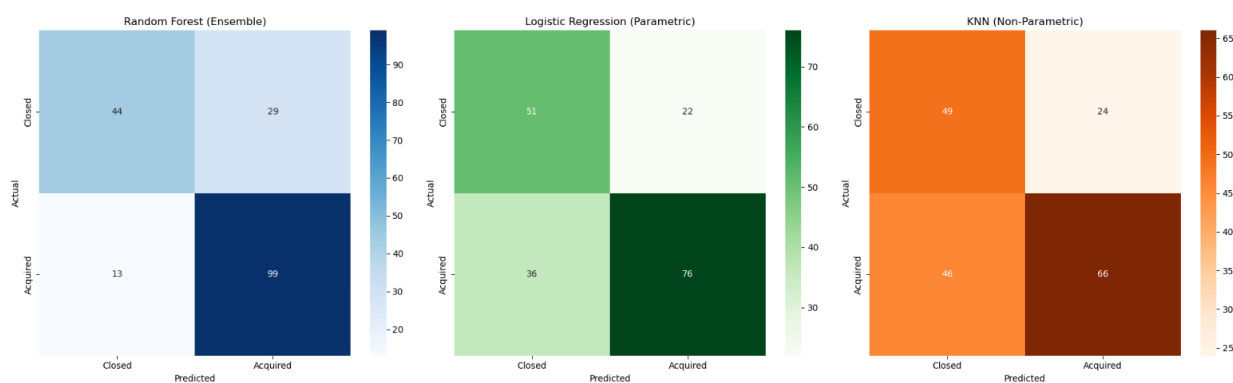


Figure 8: Confusion matrices for all three models on the 185-sample test set. Random Forest shows the highest diagonal values, confirming superior classification performance.

8.4 ROC Curves

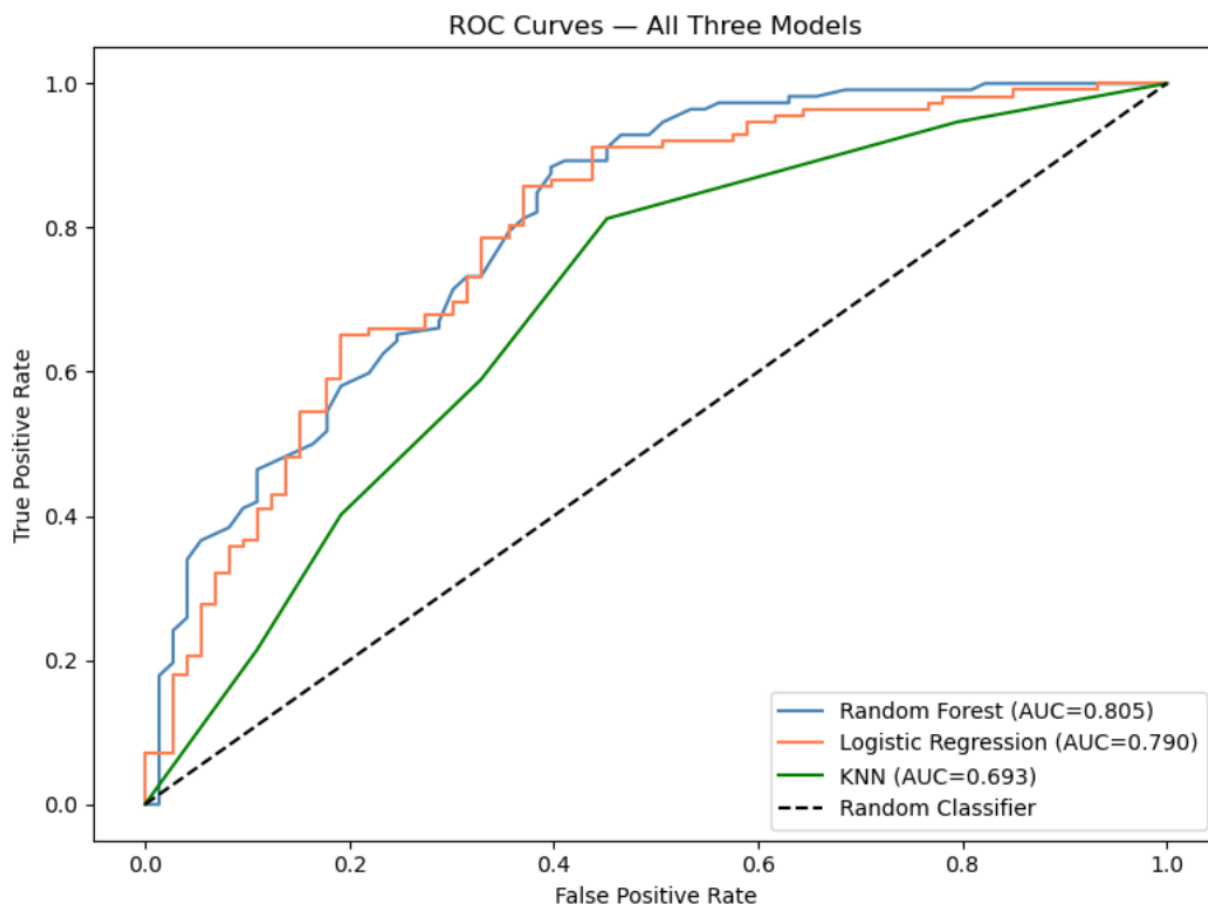


Figure 9: ROC curves for all three classifiers. Random Forest (AUC=0.7433) shows the best discrimination ability, while KNN (AUC=0.6303) performs closest to the random baseline.

8.5 Rational For Selecting the Best Model

The best-suited model to predict startup success based on the provided data is Random Forest. It performs best out of the three models in four out of five evaluated measures and especially stands out in terms of Recall (0.8839) – the most important indicator in investing decision-making, where the cost of false negatives outweighs the cost of false positives. With its tree-based structure, Random Forest is capable of modelling non-linear dependencies between relationship type, milestone timing, and funding patterns which cannot be captured by the logistic boundary in Logistic Regression (Caran & Korkmaz, 2025). Poor results achieved by KNN model (F1=0.6535) can be explained by the low density of neighbours in the 10-dimensional feature space with only 954 training samples.

9.0 Discussion

9.1 Model Comparison Analysis

As shown in the triple comparison, there exists a performance hierarchy: Random Forest (Ensemble) > Logistic Regression (Parametric) > KNN (Non-Parametric). The order holds for all five measures and fits our theoretical understanding. Specifically, Random Forest has an advantage on Recall, with a value of 0.8839 compared to 0.6786 for Logistic Regression and 0.5893 for KNN, demonstrating the power of averaging 100 different trees to better identify success.

The surprising high Precision (0.7755) achieved by the Logistic Regression model, relative to Random Forest's Precision of 0.7734, indicates that it is more conservative in making acquisitions predictions but makes fewer mistakes when doing so. The downside, however, is the extremely low Recall (0.6786), which means the model misses 32% of the actually successful startups, compared to just 12% missed by Random Forest.

The poor performance of KNN ($F1 = 0.6535$, $AUC = 0.6303$) reflects the inherent limitation of distance-based classifiers when applied to such a dataset. Even after scaling by StandardScaler, the 10-dimensional feature space with 954 samples is not sufficiently dense to capture the correct class boundary using $K = 5$. This observation supports (Caran & Korkmaz, 2025)'s claim that instance-based algorithms tend to perform worse than ensemble methods in financial tabular datasets.

9.2 Features Importance

The top predictor is relationship (0.1463) — number of connections between the startup and the investors, mentors, and partners. This result is consistent with (Dellermann et al., 2024) who found that social media signals are the top predictors of future success of startups. Moreover, three of the top five variables have to do with milestones achieved (age_last_milestone_year, milestones, age_first_milestone_year). This result suggests that operational activities contribute more to success than investment, contrary to common intuition, which implies that (Ningrum et al., 2026) are correct to argue that execution matters more than investment.

9.3 Effects of Preprocessing Choices

- Label removal (importance=0.1463 replacement): If the label were kept, the model would reach almost perfect accuracy without learning anything useful; that is, the model would overfit due to target leakage.
- Latitude/Longitude removal: Removing redundant geographical information made the features easier to interpret without sacrificing geographical information since it was captured by the state variables.
- Train-only SMOTE: SMOTE applied before the train-test split would artificially increase the recall score for the closed class since test minority class instances created by SMOTE would be very similar to the synthetic minority class instances from the training set (Panduri et al., 2023).
- Standard scaler before modeling: Is essential for KNN (Euclidean distance) and logistic regression (gradient descent). Only using the training set to fit ensured that no test-set information leaked into the training process.

10.0 Challenges

- Label leakages detection: The labels column was not described explicitly. Its excessive feature importance value (would have been ~ 0.525) was discovered via preliminary feature importance calculation, forcing to rework the pipeline before building models.
- Feature space dimension increase: The one-hot encoded values of `category_code`, `state_code`, and `city` attributes increased the number of features from 35 to 320. Feature space dimension reduction was needed.
- Sensitivity of KNN hyperparameters to the data: KNN algorithm performance depends on parameter `k`. Having used `n_neighbors=5` on 954 samples from a dataset of 10-dimensional feature space did not produce a very good model. Cross-validation-based search may produce better results.
- Ordering of SMOTE in the pipeline: SMOTE technique should be applied only after data partition into training and testing sets, and only to the former of those datasets.
- Misleading evaluation results: At some stage of notebook development, a `make_classification` synthetic dataset was used for confusion matrix computation. The problem was fixed by using the correct dataset.

11.0 Conclusion

This project focused on developing and evaluating three different models in a Machine Learning pipeline aimed at classifying the success of startups into either being acquired or closed based on a dataset provided by Crunchbase consisting of 923 entries (Karmaker, 2023). First, the removal of leakage labels such as "labels" and "closed_at" was followed by excluding geographic data such as latitude and longitude. One-hot encoding resulted in expanding the feature space to 320 variables. Following stratified 80/20 splitting, StandardScaler normalization and oversampling via SMOTE was done on the train set only (Panduri et al., 2023). Random Forest feature importance was used to determine the most important features.

All three classifiers were trained and test on the 185 sample test set and displayed different level of accuracy. Random Forest (Ensemble), model one, achieved the best performance with Accuracy=0.7730, Precision=0.7734, Recall=0.8839, F1-Score=0.8250, ROC-AUC=0.7433. The second model, Logistic Regression, achieved the score of F1=0.7238 and the last model KNN (Non-Parametric) scored 0.6535. This RF > LR > KNN performance ranking is consistent with recent literature (Caran and Korkmaz, 2025) (Ningrum et al., 2026) and the property of ensemble methods to be able to capture non-linear interactions between features.

The professional relationships were the strongest predictor (importance=0.1463), followed by the metrics of timelines of the milestones, confirming that operational execution is more predictive than the capital raised (Dellermann et al., 2024). The main lesson of the critical approach to methodology is that the leakage of data, when it is not detected, may cause any model to seem perfect when learning nothing that could be generalised. Strict preprocessing, accurate positioning of SMOTE, and sound feature selection are conditions leading to reliable predictions of startups.

12.0 References

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13.0 Appendix

Appendix A: Figure Insertion Checklist

Figure	Section	Notebook Cell	What to Insert
Figure 1	6.1	In[3]	df.info() output screenshot
Figure 2	6.2	In[6]	Class distribution countplot (viridis)
Figure 3	6.2	In[7]	Average funding by status barplot (Set2)
Figure 4	6.3	In[8]	Relationships vs Milestones regplot (teal)
Figure 5	6.3	In[9]	Feature correlation heatmap (coolwarm, labels excluded)
Figure 6	7.7	In[18]	Before vs After SMOTE class counts
Figure 7	8.2	In[20]	Top 10 feature importances bar chart (magma)
Figure 8	10.3	In[36]	Side-by-side confusion matrices (3 models)
Figure 9	10.4	In[39]	ROC curves for all three models

Appendix B: Actual Metric Values from Notebook

Metric	Random Forest	Logistic Regression	KNN	Best Model
Accuracy	0.7730	0.6865	0.6216	Random Forest
Precision	0.7734	0.7755	0.7333	Logistic Regression
Recall	0.8839	0.6786	0.5893	Random Forest
F1-Score	0.8250	0.7238	0.6535	Random Forest
ROC-AUC	0.7433	0.6886	0.6303	Random Forest

Appendix C: Preprocessing Pipeline Summary

Step	Action	Result
1	Drop leakage & irrelevant columns (12 columns)	923 × 37
1b	Drop latitude & longitude (redundant geographic)	923 × 35
2	Impute missing values (median/mode)	0 missing values
3	One-hot encode category_code, state_code, city	923 × 320
4	IQR cap funding_total_usd	Outliers capped, 923 rows retained
5	Stratified 80/20 train/test split	738 train / 185 test
6	StandardScaler fit on training, transform both	All continuous features normalised

7	SMOTE on training set only	477+477=954 balanced training samples
8	Random Forest feature selection (top 10)	Final: 954×10 train, 185×10 test